

Plantar Pressure Mapping (PPM) Spatiotemporal Framework

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Revision History

Version	Action Notes	Date	User
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Table of Contents

1. SUMMARY	3
2. REQUIREMENTS.....	3
3. GETTING STARTED	3
a. Setup Folder Structure	3
b. Generating .txt files containing the data for the framework (Figure 3)	4
4. RUNNING THE FRAMEWORK – MPP ANALYSIS	6
h. FPA computed within the framework	7
5. RUNNING THE FRAMEWORK – FRAME ANALYSIS	9
6. FINAL OUTPUTS.....	11
7. RECOMMENDATIONS.....	11

1. SUMMARY

- a. This standard operating procedure (SOP) outlines the steps to analyze dynamic barefoot plantar pressures at spatially aligned grid locations across the foot plantar surface through stance. The spatiotemporal analysis includes analyzing maximum pressure distributions and plantar pressures sampled across the stance phase of gait. This developed framework enables sensor-level statistics.
 - i. The current framework was built for Novel-EMED plantar pressure platforms but could be adapted to fit others.

2. REQUIREMENTS

- a. MATLAB 2024a or newer.
 - i. Requires the Image Processing Toolbox.
 - ii. Built as a main code [*PPM_Stance_Intervals_EMED.m* OR *PPM_Stance_Intervals_EMEDxl.m*] and one function [*resample_force_preserve.m*], for more information on the function, see section 4.j.
- b. Novel-EMED systems.
 - i. Requires the EMED-X or EMED-XL platforms.
 - ii. Software package including the ASCII software.

3. GETTING STARTED

a. Setup Folder Structure

- i. The user should create a main folder that contains all collected and analyzed spatiotemporal plantar pressure data and is titled by user choice, for this SOP example it is "PPM_DATA".
- ii. In this main folder, each study participant has their own folder containing the data.
- iii. Within each study participant folder, the data is split into raw data collected and data for analyzing with the framework in .txt file format (see section 3.b).
- iv. The raw data is saved as individual trials per foot side.
 - 1. Saved as single foot/step .dat EMED files if collected with the EMED-X (one step per trial).
 - 2. Saved as multi-step .kor EMED files if collected with the EMED-XL and extracted as a single step from the two steps collected during a single trial within the EMED data collection software (Figure 1).
 - 3. Although no specific naming convention is required, best practice would include the trial number and foot side.

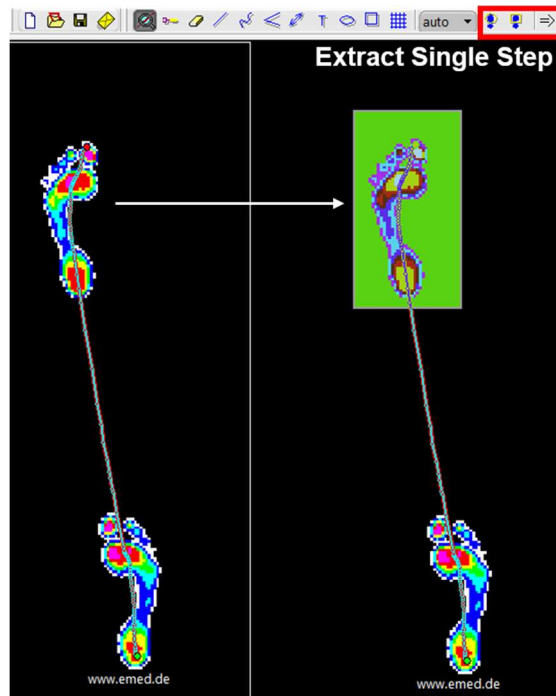


Figure 1. Following data collection with the EMED-XL platform, extracting a single step or foot side from the two steps collected in the EMED data collection software.

- v. This folder and file structure enables ease of creating .txt files for analysis with the framework and can be seen below in Figure 2.

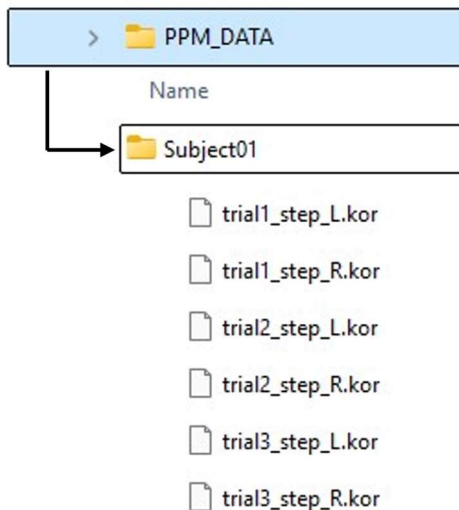
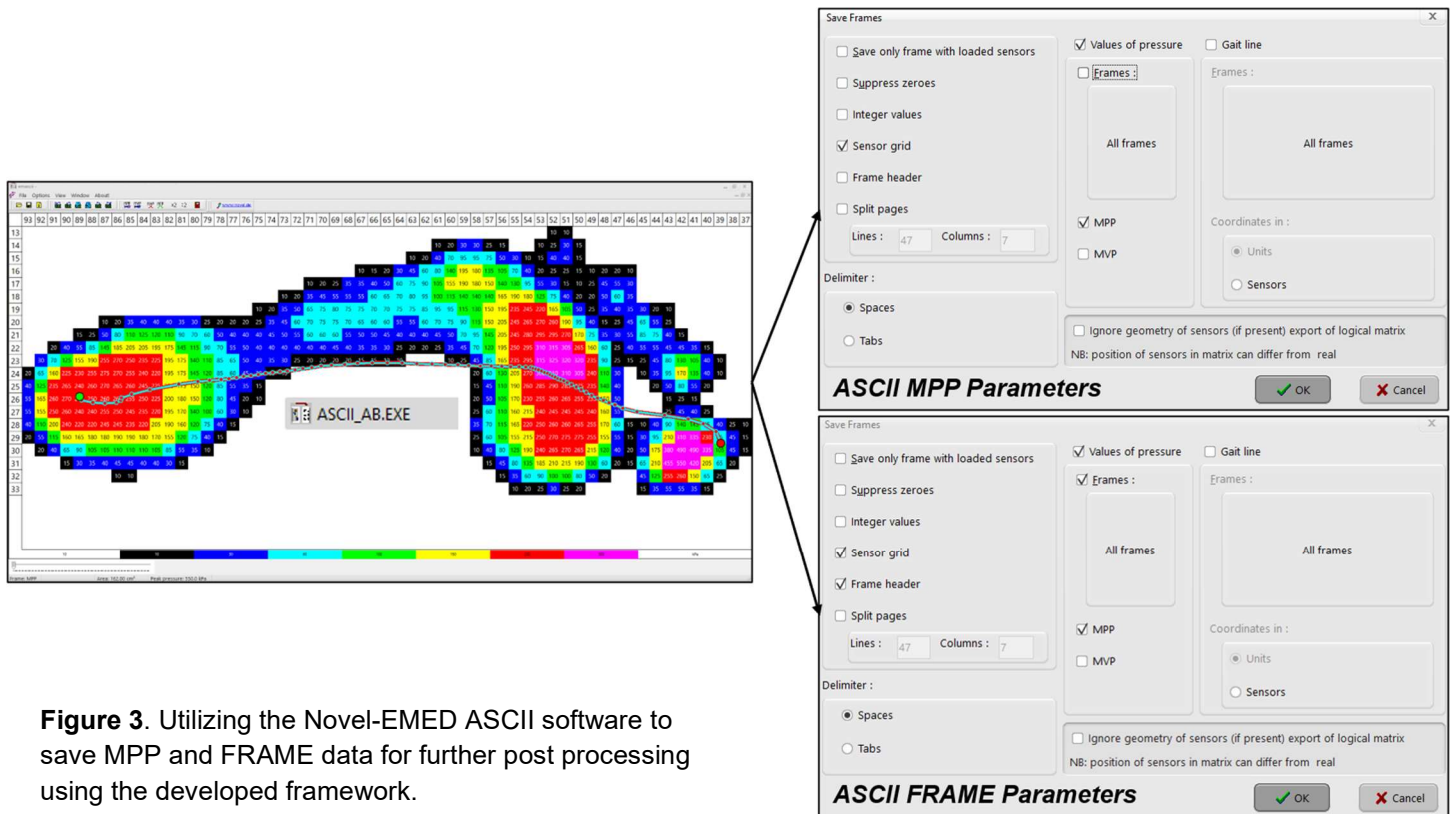


Figure 2. Folder and file structure of raw plantar pressure data for further analysis with the developed framework.

b. Generating .txt files containing the data for the framework (Figure 3)

- i. Open the ASCII Novel-EMED software and select a trial/step from above to open.
- ii. Select the save button and check the following boxes/parameters as seen for the maximum pressure picture (MPP).
- iii. Repeat for the same trial/step for the stance frames (FRAME). The individual stance frames save at 10 msec intervals.
- iv. Repeat this process for all trials/steps.



- v. All files will be automatically saved as .lst files following these steps. Open each and save as a .txt file for better compatibility with MATLAB.
- vi. When saving each individual file, please follow the following naming conventions:
 1. All MPP files should be saved as “MPP_#”, where the # corresponds to the trial.
 2. All stance FRAME files should be saved as “FRAME_#,” where the # corresponds to the trial.
 3. Best practice is to have the trial numbers range from 1 to 6, with 1 through 3 representing the left foot and 4 through 6 being the right.
- vii. The final .txt file format for analysis with the framework can be seen in Figure 4.

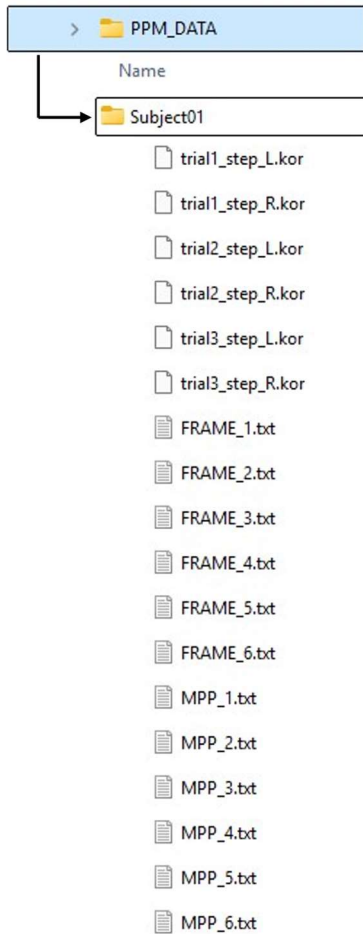


Figure 4. Folder and file structure of all data, including the .txt files for analysis with the PPM spatiotemporal framework.

4. RUNNING THE FRAMEWORK – MPP ANALYSIS

**Note: All data is processed and exported as left feet.*

- Prior to running the framework MATLAB code, make sure you have all the data and MATLAB code (folders and subfolders) in the MATLAB search path.
- In either main code [i.e., *PPM_Stance_Intervals_EMED.m*], click “Run” and follow the prompts.
- The first prompt will be to identify the file locations, please select the specific subject folder as seen in Figure 2 and Figure 4 (i.e. Subject01).
- You will then be instructed to input the trial number and select foot side, see section 3.b.vi.
- A new folder will be created with the naming convention: “Sample_trial#_FootSide” to store generated figures and processed data (see Figure 5).

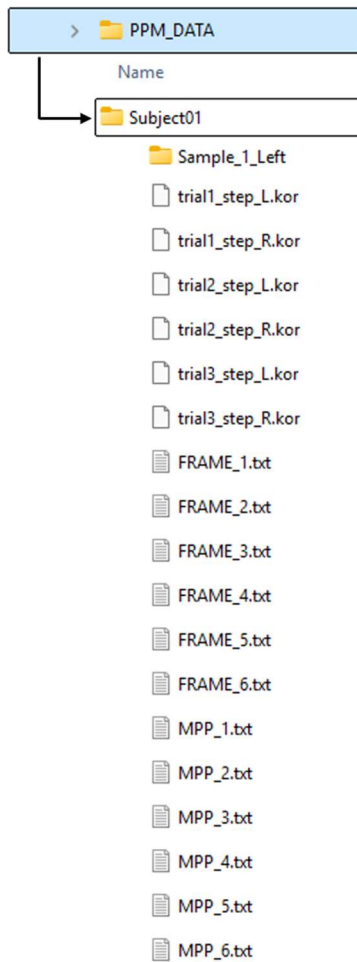


Figure 5. Folder and file structure when a sample folder including the trial number and foot side gets created for all figures and data in the framework process.

- f. The framework will import and read an MPP .txt file corresponding to the specified trial number and create a figure of the raw MPP data across the entire EMED sensor platform (see Figure 7).
- g. A contour of the MPP data will then be created for foot progression angle (FPA) computation.
- h. FPA computed within the framework**
 - i. The FPA can be imported through Novel software calculation or computed through user selection. If imported, type the number into the box provided (see Figure 6), if not, enter 0 (zero) for user definition.
 - ii. When computing the FPA, the user selects four points in the following order:
 - 1. Lateral forefoot
 - 2. Lateral hindfoot
 - 3. Medial forefoot
 - 4. Medial hindfoot
 - iii. Due to variability in toe plantar pressures, the user should focus on the ball of the foot when selecting points in the forefoot region (point 1 and 3).
 - iv. Tangent lines on the lateral and medial boundaries of the MPP contour are plotted, and the user is asked to confirm their position (see Figure 6).
 - v. The MPP will be rotated using bilinear interpolation to reduce the FPA from toe-out and toe-in to neutral.
 - vi. The entire FPA computation can be seen in Figure 6.

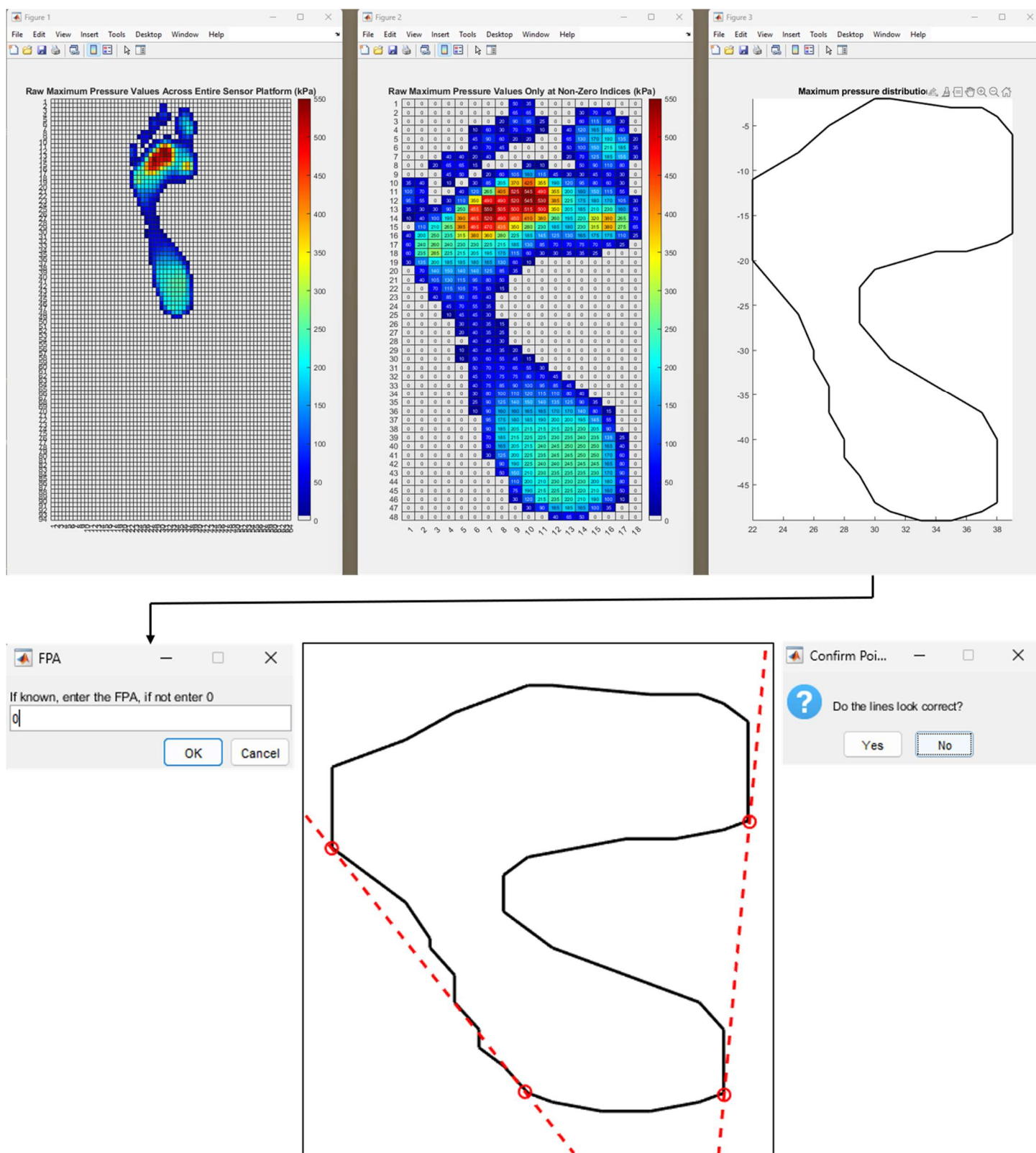


Figure 6. Extracting the MPP contour and computing the FPA based upon 4 user-selected points to create tangent lines fit to the lateral and medial boundaries for plantar pressure data rotation.

- i. With the rotated MPP data, a bounding region will enclose and reindex pressure sensors with positive values only. This data is extracted, plotted as a figure, and saved for future analysis.
- j. With the bounding region, each MPP is resampled using a force preserving approach [*resample_force_preserve.m*] so all study participant's MPP is 50 grid rows by 18 grid columns. The selection of 50x18 grid elements is user choice, 50x18 was found to be the approximate mean of control participants included during framework development.
 - i. To change the 50x18 grid, change lines 193 and 194 of the main code.
- k. The rotated, bounding region extracted, and resampled MPP data can be seen in Figure 7.

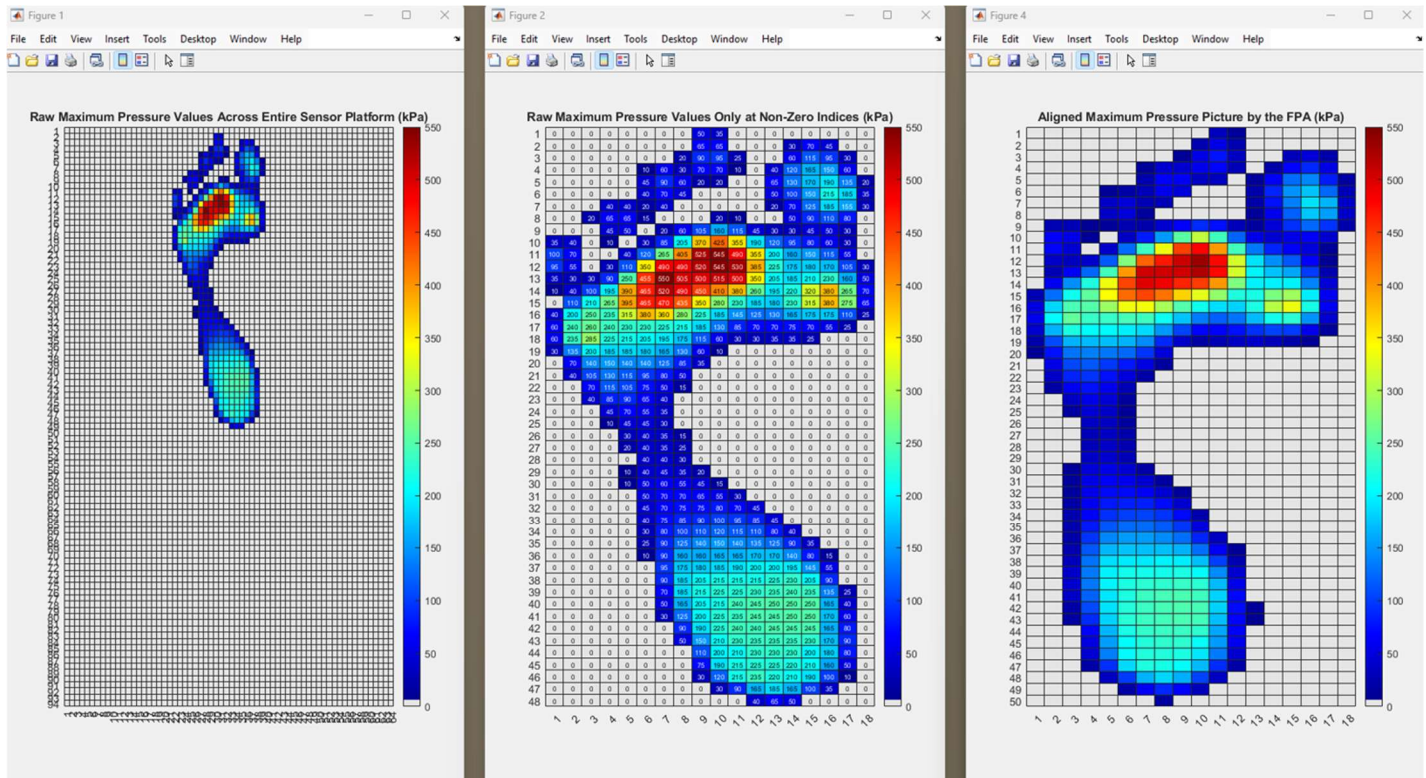


Figure 7. Raw MPP data rotated by the FPA, bounding region extracted, and force-preserved resampled to create an aligned MPP distribution for comparisons.

5. RUNNING THE FRAMEWORK – FRAME ANALYSIS

- a. The FPA and bounding region from the MPP section (section 4) are utilized in the FRAME analysis.
- b. This portion of the framework will import and read in a FRAME .txt file and compute the number of 10 msec frames included from the specific trial.
- c. The framework linearly interpolates the frames into 5% stance intervals from 0% to 100% of stance, resulting in 21 frames for analysis.
 - a. The 5% interval choice is user defined. It is recommended to choose between 2%, 5%, or 10%. To change the 5% interval choice, it is hard-coded on line 283 in the main codes. To change to 2% choose 51 TargetTimepoints and for 10% choose 11.
- d. In a MATLAB for loop, all 5% intervals are then rotated by the FPA using bilinear interpolation and data extracted using the same bounding region and force-preserved resampled (50x18 grid elements).

- e. The framework will plot the raw 5% interval pressure data across the entire platform, rotated 5% intervals, and the extracted/resampled data. An example of these plots can be seen in Figure 8.

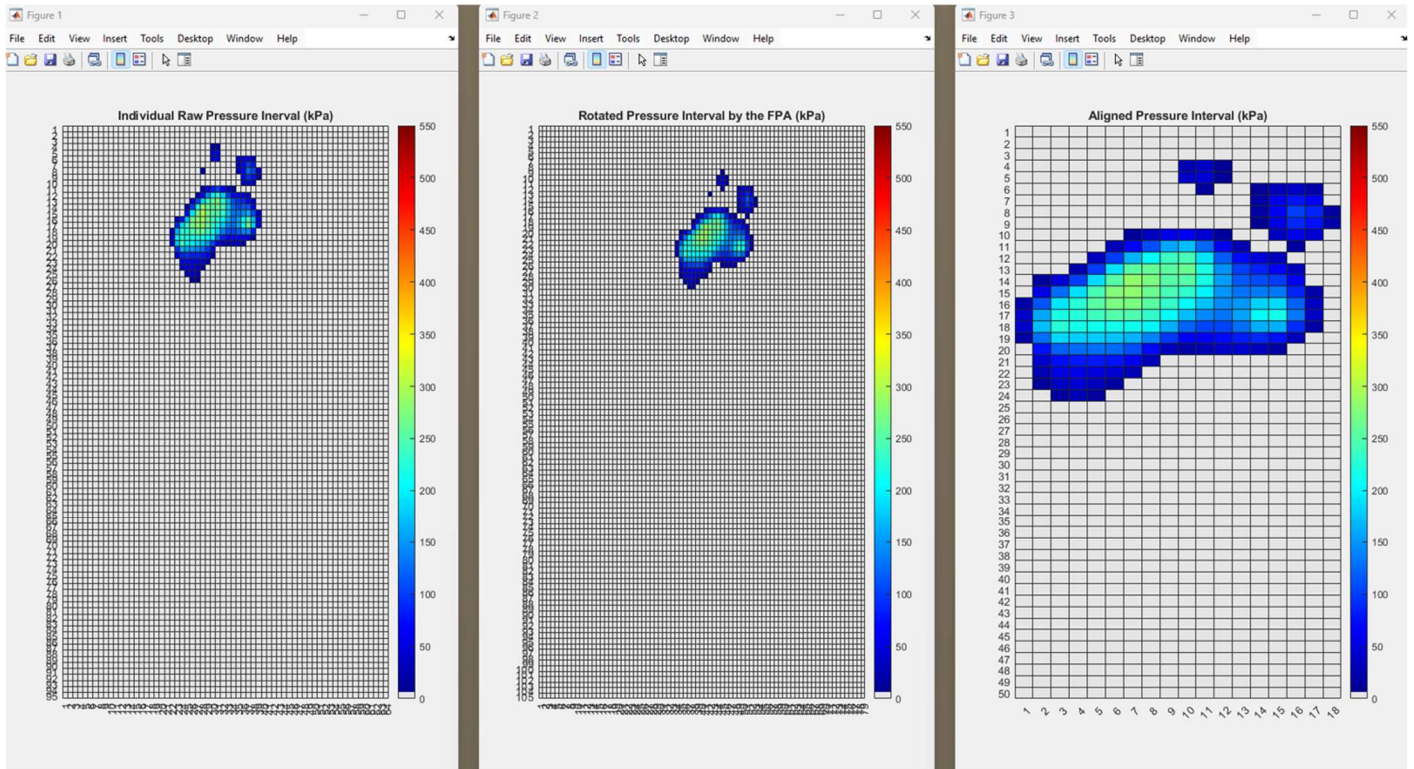


Figure 8. Raw FRAME data rotated by the FPA and resampled to create aligned 5% stance intervals (FRAME) distributions for comparisons.

- g. Following this process, a center of pressure (COP) trajectory is also calculated with 21 values, 1 per 5% stance interval (Figure 9).

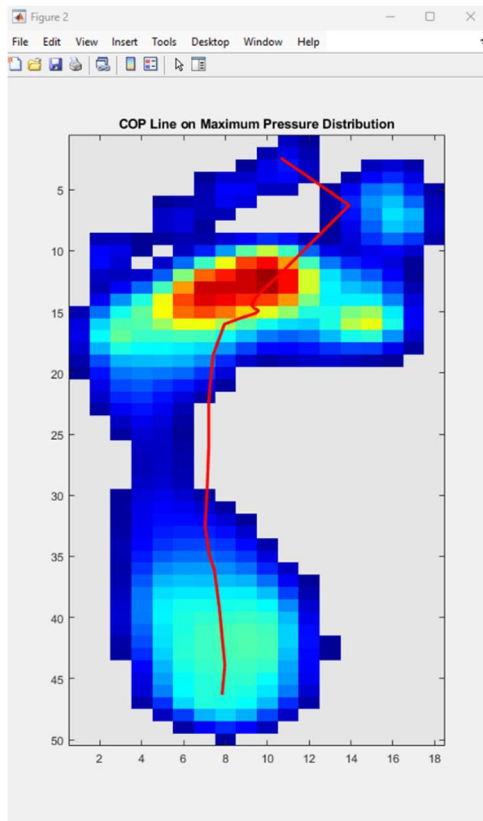


Figure 8. Aligned COP trajectory plotted on top of the aligned MPP.

- h. A “peaks preserved” analysis approach is also included in the framework with rotation and size resampling interpolation using the nearest neighbor approach. This approach was included if the user wants to focus on peak pressure values instead of the distribution of pressures.

6. FINAL OUTPUTS

- a. As MATLAB .fig file format, the raw MPP and raw frame data sampled across the entire platform are saved to the created folder. Further, the aligned MPP, 5% interval, FPA selection, and COP figures are saved to the same location.
- b. All raw and aligned data is also saved to an Excel file titled “Aligned_Pressure_Data.xlsx”.
- c. Aligned_Pressure_Data.xlsx contains the following:
 - i. Sheet1 → FPA angle, aligned MPP, raw MPP, aligned MPP using the peak preserved approach.
 - ii. Sheet2 → Peak pressure values at each stance interval using the peak preserved approach, aligned stance intervals, raw frame data
 - iii. Sheet3 → COP indices (grid locations) for each stance interval

7. RECOMMENDATIONS

- a. For collecting EMED/plantar pressure data in general, 3 to 5 trials should be collected per foot.
- b. With 3 to 5 trials, following processing with the framework as demonstrated above, the data should be averaged across all trials per foot. This would result in one aligned MPP and one aligned 5% stance interval dataset per foot.

- c. Post statistical analysis of aligned plantar pressure data should be primarily through statistical parametric mapping (SPM) for individual grid-level statistics.
- d. Prior to statistical analysis or comparisons of means among groups, normalization techniques should be considered based upon group information including BMI and walk speed during data collection.